CSCI 1315
Introduction to Proofs

Learning Objective: What is a proof, and what are some common techniques that we will see in a proof?

Key Topics: theorem, proof, definition, proposition, lemma, corollary, premise, conclusion, instantiation, generalization.

While you wait: What do you know about proofs? What are they? Why do we have them? What do they look like?

Background Terminology

A **Theorem** is a statement that can be (has been) verified as true.

A **Proof** of a theorem is a written verification that shows that the theorem is definitely true.

A **Definition** is an exactly, unambiguous explanation of a mathematical word or phrase.

A **Proposition** is a theorem that may not be as important as another theorem.

A **Lemma** is a theorem whose purpose is to prove another theorem.

A **corollary** is a result that is an immediate consequence of a theorem

Basic Outline of Proofs

Key Topics

Typically our proofs are of the form $P \Longrightarrow Q$. How do we go about solving this?

- Start with a **Premise.** You are given the proposition P, so claim that it is true.
- Build off of your premise by making logical deductions/inferences.
- Arrive at a conclusion (the Q).

Common Techniques

There are two techniques that we will find extremely useful when proving statements: **instantiation** and **generalization**.

Universal instantiation: Given the premise $\forall\, x \in U, P(x)$, we can conclude that if $c \in U$, then $P(c)$ is true (for any c).
Existential instantiation: given the premise $\exists\, x \in U, P(x)$, we can conclude that there is some $c \in U$ for which $P(c)$ is true (for some c).
Universal generalization: given the premise $P(c)$ is true for an arbitrary element $c \in U$, we can conclude that $\forall\, x \in U, P(x)$.
Existential generalization: given the premise $P(c)$ is true for some particular element $c \in U$, we can conclude that $\exists\, x \in U, P(x)$.